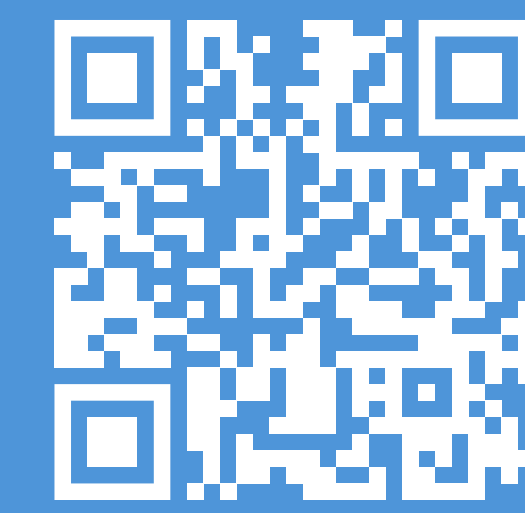


Comparing and correcting robustness metrics for quantum optimal control

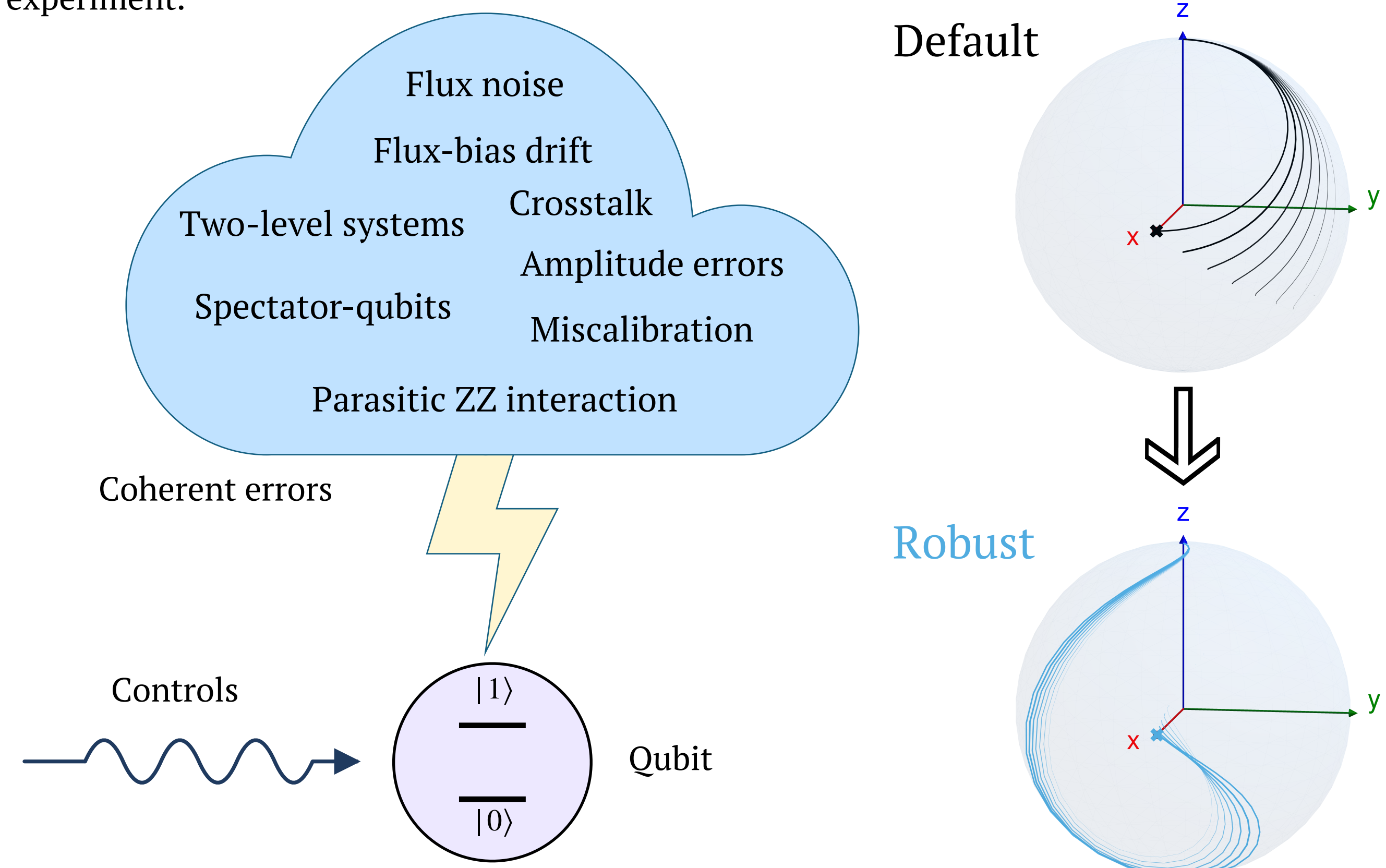
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Overview

Control pulses that optimize for fidelity are sensitive to hardware drift and modeling errors. Robust quantum optimal control [1] seeks error-insensitive pulses that maintain fidelity thresholds and obey hardware constraints. Here, we present the first systematic comparison of the two standard first-order susceptibility estimators — adjoint end-point and toggling-frame — revealing important numerical differences, and introduce a critical discretization correction to the toggling-frame estimator. Under realistic constraints, we further show that physics-informed robust controls outperform universally-robust solutions when the dominant error channels are known. Robust single-qubit gates have been demonstrated on hardware; two-qubit gates are prepared for experiment.



Robustness metrics

Consider the Hamiltonian $H(t) + \epsilon E(t)$. The ideal dynamics are given by $i\partial_t U(t) = H(t)U(t)$. But an error $E(t)$ with strength ϵ creates a perturbed evolution $\partial_t U_\epsilon(t) = -i(H(t) + \epsilon E(t))U_\epsilon(t)$. Decompose $U_\epsilon(t) := U(t)\widetilde{U}(t)$, then the interaction propagator obeys $\partial_t \widetilde{U}(t) = -i\epsilon \widetilde{H}(t)\widetilde{U}(t)$, where $\epsilon \widetilde{H}(t) = \epsilon U^\dagger(t)E(t)U(t)$. The solution is a time-ordered integral [2]

$$\widetilde{U}(t) = \mathcal{T} \exp \left(-i \int_0^t \epsilon \widetilde{H}(\tau) d\tau \right) \approx I - i\epsilon \int_0^t \widetilde{H}(\tau) d\tau + O(\epsilon^2).$$

The **first-order susceptibility** is $\mathcal{E}(E(t)) := \frac{1}{t_f^2} \left\| \int_0^{t_f} U(t)^\dagger E(t) U(t) dt \right\|^2$, where we use a scaled Frobenius norm $\|A\|^2 := \frac{1}{d} |\text{Tr}(A^\dagger A)|$.

1. The **toggling objective**: $\mathcal{E}_T^{(j)} = \frac{1}{t_f^2} \left\| \sum_{k=1}^{N-1} \Delta t U_k^\dagger E_k^{(j)} U_k \right\|^2$ is a perturbative-in- Δt discretization

of $\mathcal{E}$, with $E_k^{(j)} = \sum_{n=0}^j \frac{i^n \Delta t^n}{(n+1)!} \text{ad}_{H_k}^n(E_k)$. The conventional $j=0$ truncation drops $O(\Delta t)$ terms.

2. The **adjoint objective**: $\mathcal{E}_V := \frac{1}{t_f^2} \left\| U^\dagger(t_f) \frac{\partial U_\epsilon(t_f)}{\partial \epsilon} \Big|_{\epsilon=0} \right\|^2$ follows from the Taylor expansion of

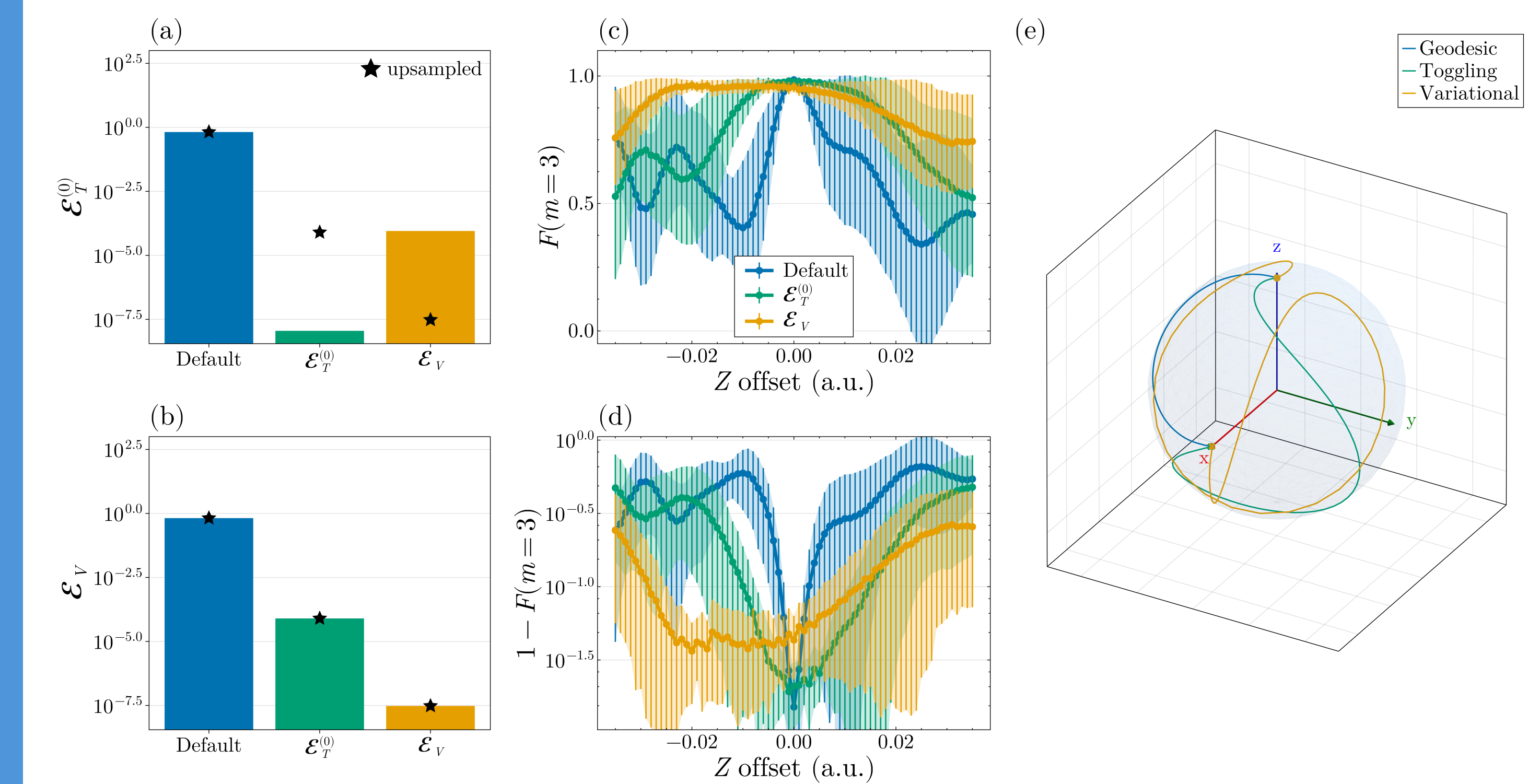
the noisy unitary $U_\epsilon(t_f) = \sum_{n=0}^{\infty} \frac{\epsilon^n}{n!} \partial_\epsilon^n U_\epsilon(t_f) \Big|_{\epsilon=0} \approx U(t_f) \left(I + \epsilon U^\dagger(t_f) \partial_\epsilon U_\epsilon \Big|_{\epsilon=0} + O(\epsilon^2) \right)$.

3. The **universal objective**: $\mathcal{E}_U := \frac{1}{t_f^2} \left\| \int_0^{t_f} U(t) \otimes U(t)^* dt \right\|^2$ can also be discretized using a

similar approach to the toggling method (see paper for more details).

Hardware Results

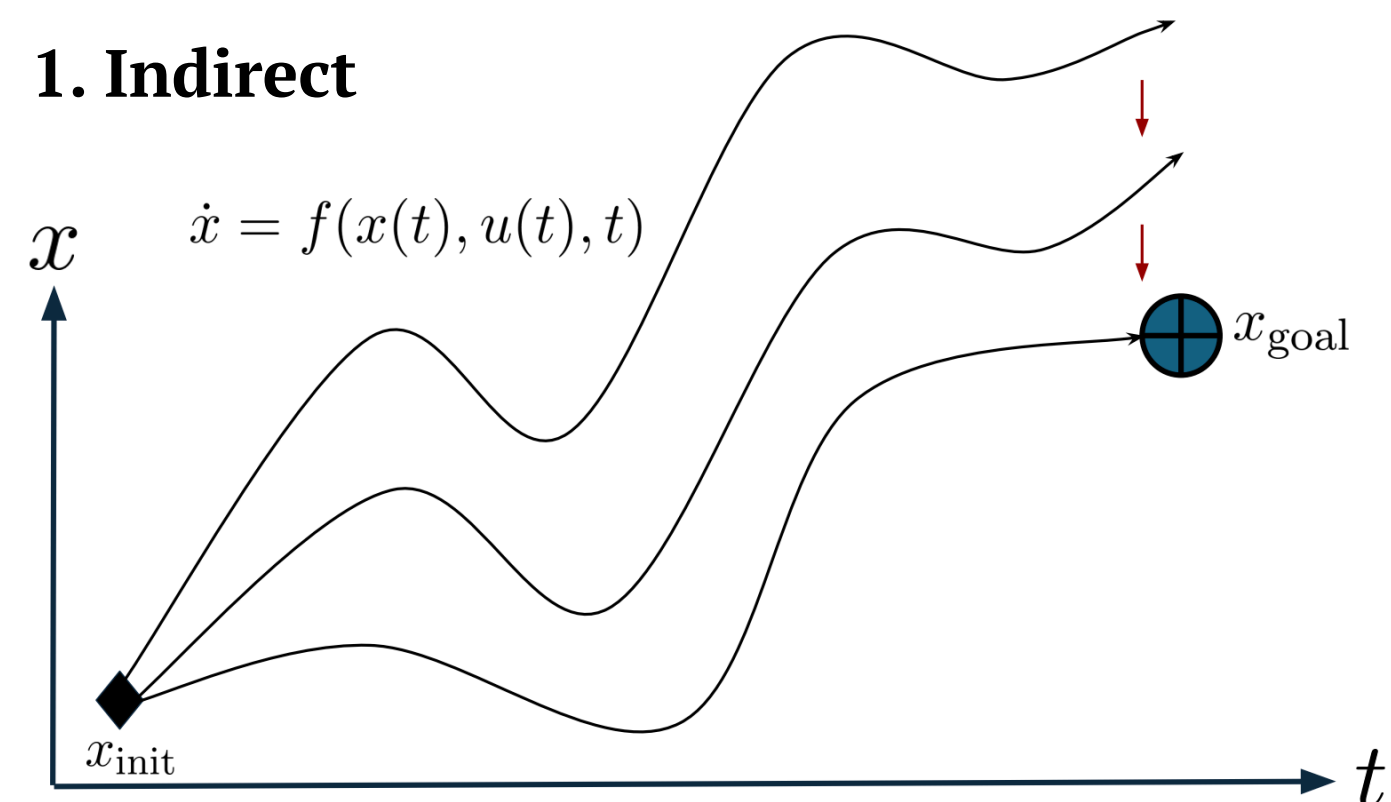
We tested three 50 ns Hadamard pulses on a flux-tunable transmon at the Cleland group. The pulses were optimized for fidelity only (default, blue), the zeroth-order toggling objective (green), and the adjoint objective (orange). (a) Zeroth-order toggling susceptibility for the three pulses from numerical simulation, with stars indicating the ground-truth susceptibility computed with the controls upsampled to 2^{16} time steps. (b) Adjoint susceptibility for the same three pulses from numerical simulation, with stars indicating the ground-truth susceptibility. (c) Sequence fidelity from randomized benchmarking with depth $m=3$, averaged over 10 trials, as a function of a static shift in qubit frequency. (d) Bloch trajectories for the three pulses. The zeroth-order toggling metric in (a) is not consistent with the ground-truth susceptibility (stars) or with the experimental ordering of robustness in (c), while the adjoint metric (b) agrees with both.



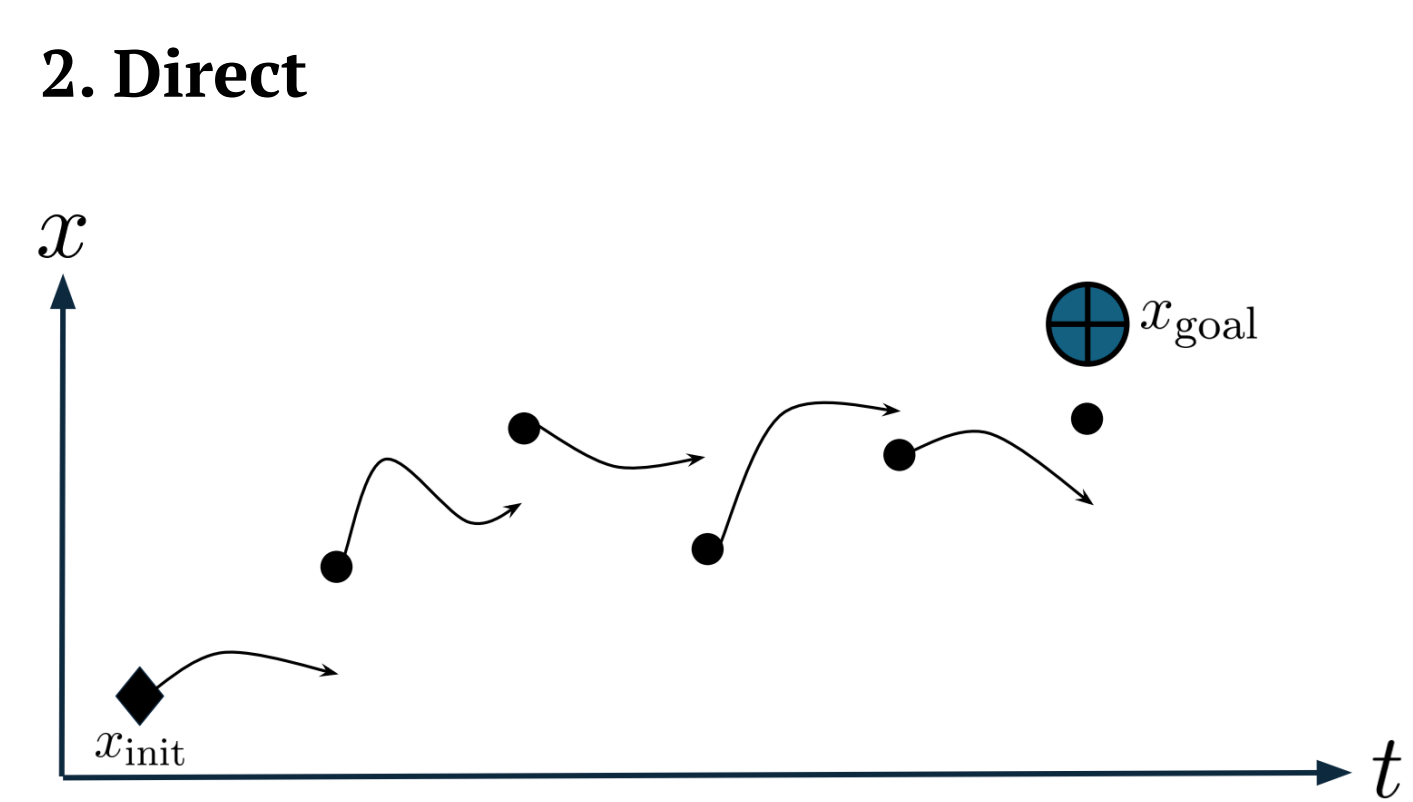
Quantum Optimal Control

Quantum optimal control casts pulse design as a trajectory optimization problem: states $\mathbf{x}(t)$ and controls $\mathbf{u}(t)$ are discretized into knot points, and an objective is minimized subject to the system dynamics f — the Schrödinger equation for closed systems or the corresponding master equation for open systems. For simplicity, we specialize to the Schrödinger equation with controls subject to hardware-imposed constraints. **Indirect methods** optimize over controls alone, enforcing dynamics implicitly via rollout; **direct methods** retain states as decision variables and enforce dynamics as constraints between sequential knot points — and converge more reliably under constraints (bottom right).

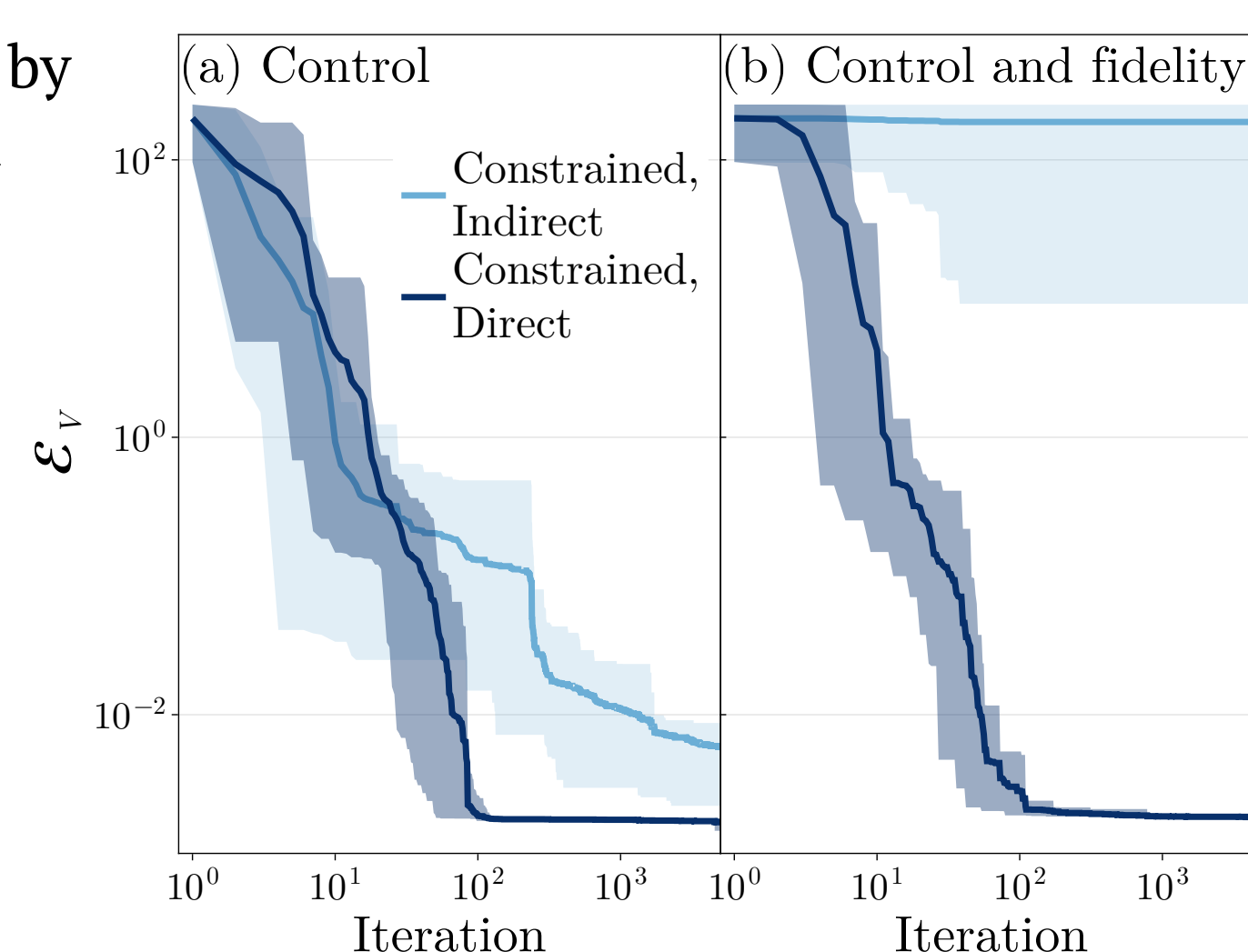
1. Indirect



2. Direct

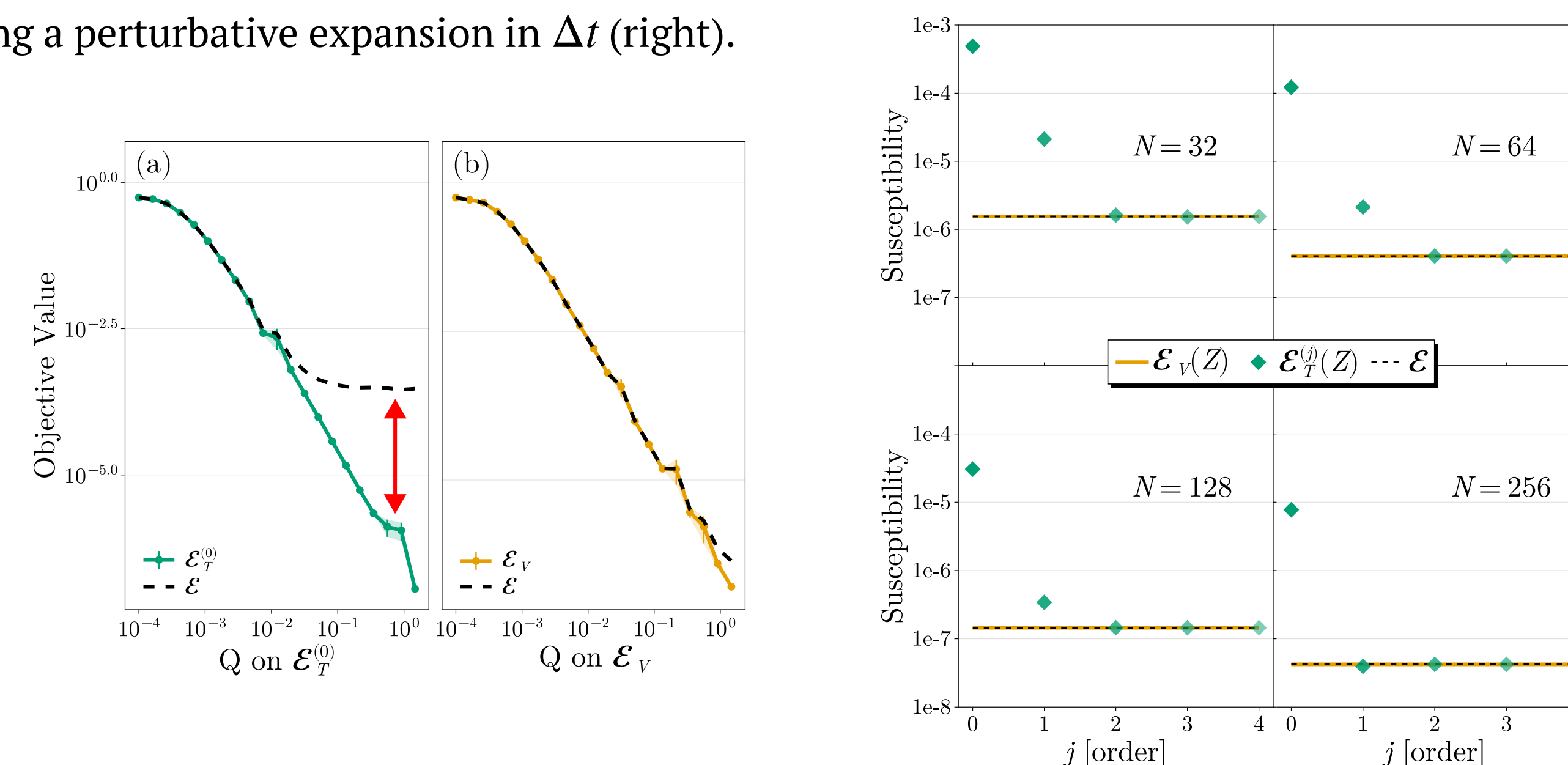


We rely on a package called Piccolo.jl provided by Harmoniqs — an open-source quantum control software leveraging techniques from robotics.

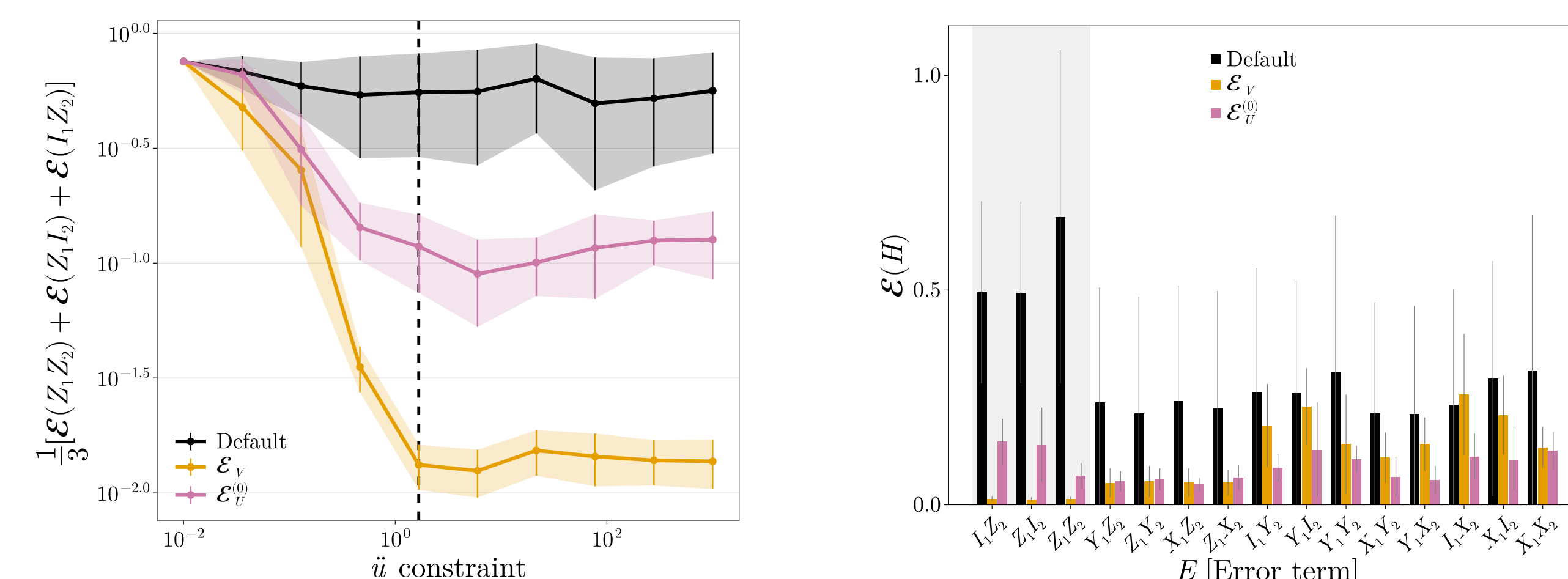


Numerical Experiments

The zeroth-order toggling objective over-exploits a discretization error, minimizing the inaccurate metric rather than the actual susceptibility (left). We repair the discretization error using a perturbative expansion in Δt (right).

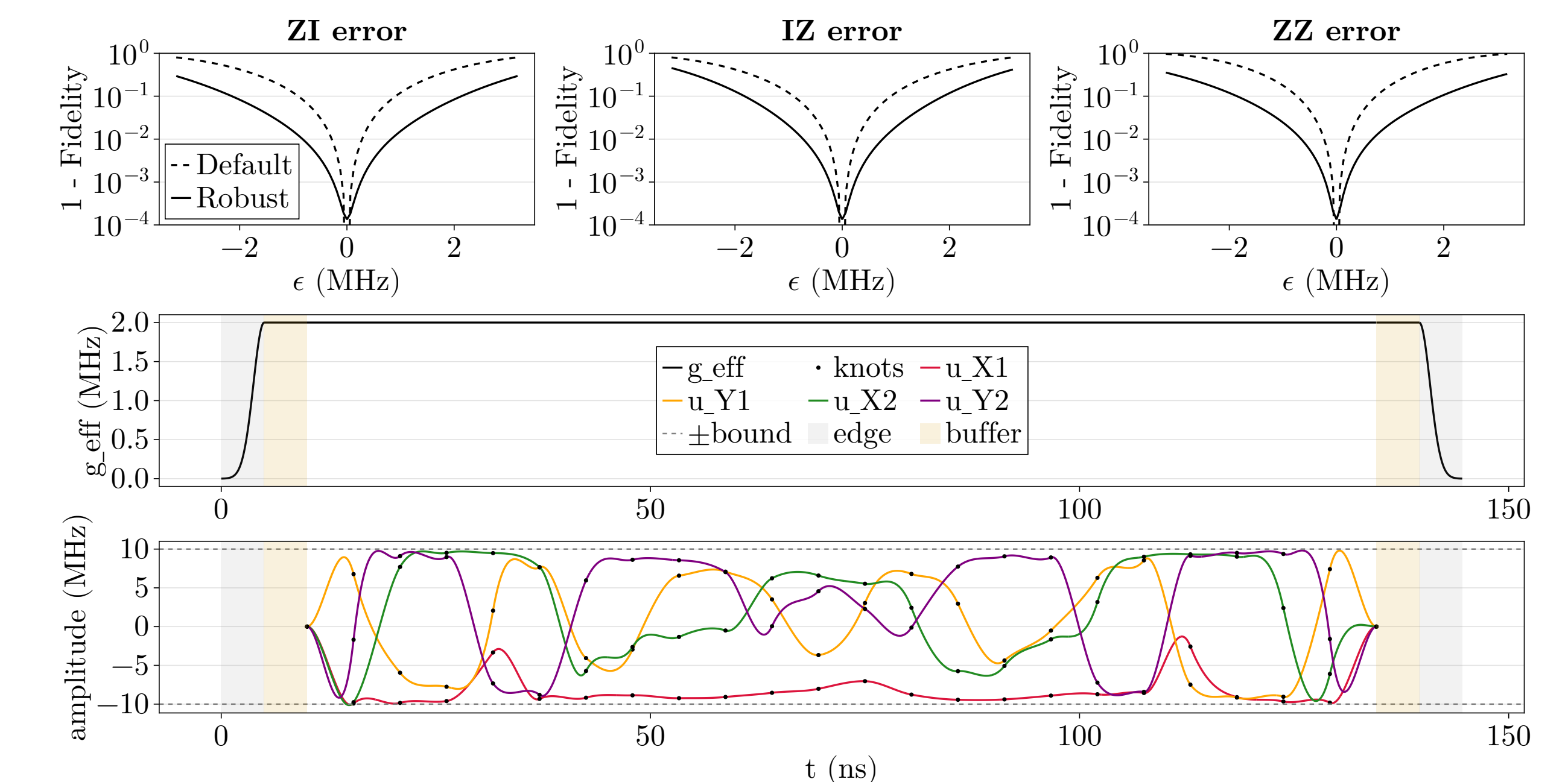


We find that for constrained problems, targeting specific Pauli error channels (adjoint, orange) yields significantly lower susceptibility on those respective channels than the universal approach (pink), which must distribute robustness across all errors.



Future Work

- **Robust two-qubit gates.** Develop an experimentally feasible robust two-qubit gate using square flux drives and concurrent single-qubit microwave drives (see robust iSWAP below).
- **Time-dependent errors.** Extend the adjoint method beyond quasi-static errors using noise spectra measured on hardware.
- **Closed-loop calibration.** Combine open-loop robust controls with sample-efficient hardware calibration to enable dynamic flux control.



Citations & Acknowledgments

Citations: [1] Propson, T., et al. *Phys. Rev. Applied* **17**, 014036 (2022). [2] Trowbridge, A., et al. *IEEE QCE* 2023, vol. 1, 1278 (2023). [3] Sakurai & Napolitano, *Modern Quantum Mechanics*, 3rd ed. (Cambridge University Press, 2020). [4] Poggi, P. M., et al. *Phys. Rev. Lett.* **132**, 193801 (2024).

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